Mock Exam

Technische Hochschule Ingolstadt

Examination semester: Summer semester 2027
Study programme: User Experience Design, UXD (M.Sc.)
Course: Multimodal, Interactive Systems (UXDM_PMIS)
Lecturer: Prof. Dr. Simon Nestler

Name	Student ID	Date

Notes

Instructions

- The working time is 90 minutes; the maximum score is a total of 100 points (100 points in 90 minutes, so plan slightly less than a minute per point).
- Please answer each question directly in the answer box provided for it.
- The space provided for each answer has been sized according to the expected length of your answer.
- The exam tests judgment, diagnosis, and justification on cases and artifacts. There are no definition questions. Name the cost of every decision you make.
- **Please do not use any aids (AI tools, ChatGPT, Google, notes, the solution, etc.). AI is excluded in the real exam — working without tools is the only way to get a realistic picture of where you stand.**

– please fill in on your own (during the review session) –

Points	Percent	Grade

By signing with my name (in block letters) I confirm that I have assessed myself fairly and realistically in this self-assessment.

Task 1 (11P): A car warns the driver of the same obstacle simultaneously through a warning tone, a symbol in the head-up display, and seat vibration.

a) Is this redundancy or conflict? Decide, and justify your decision with the criteria from the modality framework, not with gut feeling. (7P)

b) State the condition under which your answer would flip. (4P)

Task 2 (17P): A visitor check-in terminal in a hospital corridor: the audience is aged 8 to 88, the noise level is moderate, and the data entered is privacy-sensitive. Proposal A: speech-first with touch fallback. Proposal B: touch-first with a read-aloud function.

a) Choose one proposal. Justify your choice with two context factors from the case. (9P)

b) Name one concrete disadvantage your choice buys you. (5P)

c) Name one situation in which the rejected proposal would have been the better choice. (3P)

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Task 3 (11P): An in-air swipe gesture keeps firing while users gesticulate during conversation.

a) Which parameter of the gesture's state model is missing? Name it and justify your answer. (7P)

b) State what the fix costs the user. (4P)

Task 4 (17P): The excerpt below comes from a dialog flow for a pharmacy prescription-refill voice assistant. It was generated by a language model, including its error handling; the model's design rationale is given in quotation marks. Two of its design claims are wrong.

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S1  Greeting: "Hi! I can refill your prescription.
    Which medication do you need?"
S2  The user names the medication. The assistant places the
    order with an implicit confirmation: "Ordering ibuprofen
    800 — anything else?"
    ("Implicit confirmation keeps the dialog short and
    efficient.")
S3  While the assistant is speaking, user speech is ignored
    until the prompt has finished.
    ("Ignoring interruptions keeps the dialog predictable.")
S4  If the medication is not recognized twice in a row, the
    assistant offers to connect the caller to pharmacy staff.
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a) Identify the two wrong design claims and explain, for each, why it is wrong in this context. (8P)

b) For each of the two claims, name the test or knowledge that would have exposed the error before shipping. (9P)

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Task 5 (13P): A municipal utility (Stadtwerke) wants a digital way for customers to report a move to a new address.

a) Decide: chat interface, form, or hybrid. Your justification must use one interaction-cost argument and one error-consequence argument. (10P)

b) Name the price of the hybrid, regardless of whether you chose it. (3P)

Task 6 (14P): Two published figures on automated accessibility testing: Deque reports that axe finds "57 %" — of issue *volume*, measured by the tool's maker. The GOV.UK test found that the best single tool detected about 40 % of 143 deliberately built barriers.

a) Explain why both figures can be true at the same time. (7P)

b) Which of the two figures will a checker vendor quote in marketing, and why? (4P)

c) A team must demonstrate conformity under the EAA/BFSG. What does a clean checker report prove, and what does it not prove? (3P)

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Task 7 (17P): The following design detail of a public ticket terminal is WCAG-compliant: for blind users, a tactile button starts an audio walkthrough that reads the entire screen content aloud once, in a fixed order. All input remains on the flat touchscreen; the audio describes the layout ("the green button in the lower right corner").

a) Explain why this detail still excludes blind users, even though it is compliant. (6P)

b) Propose one modality substitution or addition that removes the barrier. Be concrete: which channel replaces or complements which. (7P)

c) Name the side effect your fix has for another named user group. Who pays for it? (4P)